

Evaluation of a Multi-Agent System approach to airline disruption management

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ABSTRACT

Each day, airlines face disturbances that disrupt their carefully planned operations. Events like adverse weather conditions, sick crew members, or damaged aircraft often result in delays in the airline's schedule. An airline recovers from such disruptions through the role played by its Airline Operations Control (AOC). A Multi-Agent System (MAS) approach to airline disruption management was recently proposed under the acronym MASDIMA (Multi-Agent System for Disruption Management in AOC). The purpose of this paper is to evaluate this MAS supported AOC approach on its performance and its practical introduction. This is done using a scenario-based analysis to compare the MAS supported policy to human-team based AOC policies. A task-based analysis identifies how well AOC is able to cover a set of tasks using the MAS supported policy. The scenario-based analysis shows that the MAS supported AOC is able to find the optimal solution, and to do this significantly faster. The task-based analysis identified two main challenges for implementing the MAS supported AOC policy: i) to overcome the loss of experience that is caused by significantly automating humans roles in AOC, and ii) to reduce the workload for people that remain in AOC after its introduction. The paper concludes that implementing the MAS supported AOC policy leads to both better and faster resolutions, though the replacement of human roles also poses novel challenges that remain to be resolved: a potential increase in workload for the remaining human role and loss of experience in handling exceptional situations.

1. Introduction

Airlines constantly face disturbances that disrupt their carefully planned operations. Events like adverse weather conditions, sick crewmembers, or damaged aircraft often cause delays in the airline's schedule. Each airline has its Airline Operations Control (AOC) monitoring operations worldwide, and managing recovery from disruptions. For an airline such disruptions are very costly because they tend to cause domino effects in the highly optimized air transportation schedule. Over the year 2007 alone, U.S. carriers lost over \$8 billion because of delays of some sort (Barnhart, 2009). Reducing the impact of disruptions on the airline schedule could considerably reduce these costs. Most research on the improvement of AOC decision making policy focusses on using optimization techniques for the development of decision support tools. For instance, Bratu and Barnhart (2006) propose two optimization tools that generate recovery plans for aircraft, crews, and passengers by determining which flight leg departures to postpone and which to cancel. Abdelghany et al. (2008) propose a decision-support tool that provides AOC centres with the capability to develop a proactive schedule recovery plan that integrates all flight resources.

The optimization tool examines possible resource swapping and flight rebooking to generate a schedule recovery that minimizes flight delays and cancellations. Petersen et al. (2012) propose a mixed-integer programming tool to solve the fully integrated airline recovery problem including the schedule, aircraft, crew, and passenger problems. In the same vein, Arikan et al. (2017) propose an optimization tool to solve the fully integrated airline recovery problem using a conic quadratic mixed integer programming formulation. Santos et al. (2017), present an integer linear programming tool to help AOC controllers decide which flights to delay and which flights to make depart on time.

From a combinatorial optimization perspective, these tools have the mathematical capability in minimizing airline operating costs and passenger costs. However, such a combinatorial optimization approach fails in capturing the complex socio-technical nature of AOC (Feigh and Pritchett, 2010; Bruce, 2011a; Richters et al., 2017). In order to address these socio-technical challenges, Castro (2013) has taken a Multi-Agent System (MAS) based approach to the development of a novel decision support tool for airline disruption management. The resulting tool is referred to as MASDIMA (MAS for Disruption Management in AOC). In order to realize a better handling of the complexity of the airline

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disruption management problem, [Castro \(2013\)](#) proposes replacing several human roles in AOC by software agents. The latter implies a large change in the established AOC policy that is largely depending on coordination and decision-making by a team of humans.

Inherent to the complex socio-technical nature of airline operations, it involves dynamic interactions between multiple actors, systems and uncertainties. This makes performance evaluation and comparison of a novel AOC policy far from trivial. The challenge is twofold: i) How to identify a challenging airline disruption scenario that can be used as a benchmark? and ii) How to evaluate the application of an AOC policy on a benchmark scenario. The former problem has been addressed in [Bruce \(2011b\)](#) as follows. Supported by three experts with extensive AOC background, [Bruce \(2011b\)](#) developed a challenging airline disruption scenario, while receiving critical feedback from four other AOC experts. The major roles given to the AOC experts was to ensure that the scenario was representative of realistic and sufficiently complex disruptions ([Bruce, 2011b](#)).

[Bouarfa et al. \(2016\)](#) have shown that the second problem, i.e. to evaluate the application of an AOC policy to a given airline disruption scenario, can be addressed using Agent-Based Modelling and Simulation (ABMS). For applications in other domains, ABMS has shown its effective use in analyzing complex socio-technical systems ([Macal and North, 2010](#); [van Dam et al., 2013](#)). [Bouarfa et al. \(2016\)](#) have applied ABMS to the evaluation of the performance of four human-team AOC policies to the challenging airline disruption scenario of ([Bruce, 2011b](#)). Three of the four policies were based on current airline practice, whereas the fourth was based on joint activity coordination theory for human teams ([Klein et al., 2005](#)). For each of these four policies, [Bouarfa et al. \(2016\)](#) capture each human in the AOC team, including its specified role, as an agent in the ABMS. The obtained results in ([Bouarfa et al., 2016](#)) showed that the three current policies led to similar resolutions of the disruption benchmark as the best one identified in ([Bruce, 2011b](#)). However, the fourth policy led to a significantly better resolution of the benchmark disruption scenario. A key contribution of the current paper is to evaluate the MAS-supported AOC policy of [Castro \(2013\)](#) on the benchmark disruption scenario of [Bruce \(2011b\)](#), and to compare the results with those obtained by [Bouarfa et al. \(2016\)](#) for the other AOC policies.

The airline disruption scenario development by [Bruce \(2011b\)](#) was focused on human-based AOC disruption management policies. However, one should be aware that there may be airline disruption scenarios which are easier to resolve by a human team based airline disruption management policy, than it is for a MAS supported policy. In order to realize a better understanding of the specific types of scenarios where this might be an issue, the current paper also conducts an expert-based evaluation of remaining human tasks in a MAS supported AOC policy. This leads to the identification of scenarios that are potentially more critical for a MAS supported AOC policy.

This paper is organized as follows Section 2 provides background on both AOC and the agent-based paradigm Section 3 provides a summary of multi-agent coordination approaches from literature Section 4 describes the MAS supported AOC policy and its application in AOC Section 5 compares the MAS supported AOC policy versus the human coordination policies that have been studied in ([Bouarfa et al., 2016](#)). Section 6 describes an expert-based evaluation of the remaining human tasks in the MAS supported AOC policy. Finally, Section 7 presents the conclusions and recommendations of this research.

2. Background

2.1. Airline Operations Control

The idea of monitoring and controlling a transport network in real time is not new. The concept was first established in the 19th century in the railway industry, when the development of the telegraph made it possible for information to travel faster than physical transport ([Peters,](#)

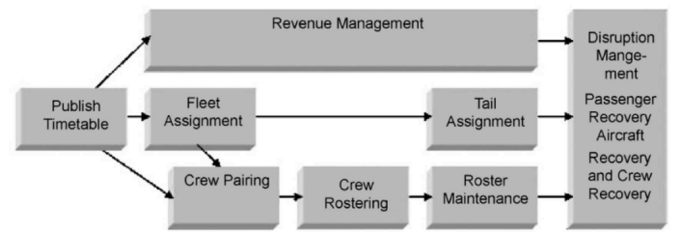


Fig. 1. Disruption management starts when airline planning ends ([Kohl et al., 2007](#)).

2006). This allowed for a central location in which real-time information about the current status of the network could be collected and acted upon. Today, the concept of monitoring operations in real-time is used across industries, with AOC as one example.

Airline disruption management starts when airline planning ends ([Fig. 1](#)). The scheduling process starts with publishing a preliminary timetable up to 1 year before the day of operations. The timetable provides the basis for the aircraft schedule, which assigns an aircraft type to each flight. With the flights and aircraft types known, crew pairing defines the amount and type of crew per flight. The next step is to assign specific aircraft and individual crewmembers to each flight in the tail assignment and crew rostering phase. After publishing the crew roster, crewmembers can request changes in their schedule in the roster maintenance phase. Disruption management starts after the airline planning process ends and is considered a tactical step during recovery ([Grandeau, 1995](#); [Clarke, 1998](#); [Kohl et al., 2007](#); [Clausen et al., 2010](#)).

During the day of operations, the airline schedule is subject to many disruptions. The four main airline schedule disruptors are aircraft mechanical problems, severe weather, airport congestion, and industrial action (e.g. strikes). The goal of AOC is to deliver customer promise despite these disruptions. In doing so, it should minimize airline costs incurred during recovery, and return to the original schedule as soon as possible ([Kohl et al., 2007](#)).

Disruptions affect the aircraft, crew, and passenger resources of an airline. Managing these resources is the duty of AOC operators. Each AOC operator has his own role. Such roles might vary per airline, but six are common to most airlines: flight dispatch, aircraft control, crew tracking, aircraft engineering, customer service, and Air Traffic Control (ATC) coordination ([Kohl et al., 2007](#)). Because the airline operations supervisor is ultimately responsible for AOC operations ([Clarke, 1998](#)), he/she has the authority to make changes in the nominal schedule.

An airline controller can manage a disruption in many different ways. To resolve a problem that affects the aircraft resource, a flight can be delayed, cancelled, rerouted, or the aircraft exchanged. Crew related problems can also be resolved by cancelling or delaying the flight, or by calling in new crew or reassigning existing crew. To resolve a passenger problem, an operations controller might change the passenger's flight or delay the passenger ([Barnhart, 2009](#); [Castro, 2013](#)).

How well disruptions are managed depends on how AOC is organized. For example in Europe, AOC often performs the task of flight following, while flight planning and dispatch is often performed outside AOC ([Kohl et al., 2007](#)), whereas in North America, flight dispatchers and planners are assumed to make an integral part of AOC ([Pujet and Feron, 1998](#); [Clarke, 1998](#); [Castro and Oliveira, 2011](#)). In the current paper we adopt the latter, which is also in line with [Bruce \(2011b\)](#), [Castro \(2013\)](#) and ([Bouarfa et al., 2016](#)).

According to [Castro \(2008\)](#) and [Machado \(2010\)](#), there are three types of AOC centers. A decision center, a hub control center, and an integrated control center. In a decision center, airline controllers are located in the same space while other functional groups such as maintenance services and crew control are located in a different physical space. A hub control center oversees the activities at the hub, which may include ground and passenger services, but other operations such as aircraft control are monitored from a different location. An

integrated airline operational control center integrates all functional groups under the same physical location. The research presented in this paper considers an integrated control center.

Work practice differs from airline to airline and from individual to individual. Smaller airlines tend to use schedule visualization software to easily enable their controllers detect irregularities, while major airlines use software that is able to automatically detect these irregularities. Operators with similar roles sit next to each other to easily communicate and collaborate. Each desk keeps the necessary communication equipment such as phone and telex. Centrally placed screens show live news, as well as weather reports and performance indicators. Clocks indicate the time in different time zones around the world (Feigh, 2008).

2.2. Agent-based paradigm

There are two main agent-based paradigms in the literature. The first paradigm is Multi-Agent Systems (MAS), and the second paradigm is Agent-based Modelling and Simulation (ABMS) of socio-technical systems. Although there is significant knowledge overlap (e.g. both use distributed autonomous agents) the two are used in complementary ways.

The primary goal in ABMS of socio-technical systems is to search for explanatory insights into the collective behavior of agents obeying specific rules (Nikolic and Kasmire, 2013; Wikipedia, 2017). The primary goal of MAS is to exploit the agent-based paradigm as an approach in resolving complex practical or engineering problems (Wikipedia, 2017). Researchers in ABMS of socio-technical systems develop simulations that can reveal system behavior emerging from the agent's collective actions and interactions. In these simulations, the agent entities are used to represent actors in the real world (e.g. individuals or teams as well as (intelligent) technical system agents. They are programmed to react to the computational environment in which they are located, where this environment is a model of the real environment in which the actors operate (Gilbert, 2008). So with ABMS comes the need to model human behavior and social interactions.

On the other hand, a technical MAS is a computerized system composed of multiple interacting (intelligent) agents. Here agents can for example conduct some methodical, procedural or algorithmic search. The main difference between ABMS and MAS is that ABMS sets up agents believed to have crucial characteristics of real world analogs to see what happens when they do whatever they do; while in a MAS agents are set up with exactly the characteristics, connections and choices that they need to achieve certain desired emergent states.

3. Multi-agent coordination approaches

This section gives an overview of multi-agent coordination approaches in software agent systems, followed by a review of complementary coordination approaches in human teams.

3.1. Coordination by software agents

One of the classic coordination approaches is the **master/slave** technique that is typically used for task and resource allocation among slave agents by a master agent (Nwana et al., 1996). The master agent plans and distributes fragments of the plans to the slaves. The slaves may or may not communicate among themselves, but must ultimately report their results to the master agent. Another classic coordination technique is the **contract net protocol** (Bourne et al., 2001). In this approach, agents assume two roles: 1) A manager who breaks a problem into sub-problems and searches for contractors to solve them, as well as to monitor the problem's overall solution, and 2) A contractor who does a sub-task. However, contractors may recursively become managers and further decompose the sub-task and sub-contract them to other agents.

Other coordination approaches include, multi-agent planning (Nwana et al., 1996), negotiation protocols (Sycara, 1989; Bussmann and Muller, 1992), and voting methods (Bosse et al., 2006). In **multi-agent planning**, agents build and maintain a multi-agent plan that details all of the future actions and interactions required to achieve their goals, and furthermore interleave execution with more planning and re-planning. Due to the re-planning feature, multi-agent planning is particularly useful in dynamic situations. **Negotiation** is defined by Bussmann and Muller (1992) as the communication process of a group of agents in order to reach a mutually accepted agreement on some matter. Sycara (1989) has explained that to negotiate effectively, agents must reason about beliefs, desires, and other agents. **Voting methods** refer to various techniques that are used to describe decision-making processes involving multiple agents. Although originating from political science, they are currently used within a number of domains such as gaming theory and pattern recognition.

The various coordination approaches presented have their relative advantages and disadvantages and there is no universally best method. In general, the theoretical methods produce good results for narrowly defined coordination problems but many of their underpinning assumptions have limitations in developing real-world systems (Lesser, 2014).

3.2. Complementary approaches in human teams

Various complementary coordination approaches are of use in human teams, ranging from routine and psychological approaches, to ecological, socio-technical and integrative approaches; i.e. a fusion of multiple different approaches (Paris et al., 2000).

Thompson (1967) identified two basic complementary coordination approaches in human teams, namely **routines/protocols** and **mutual adjustment**. The first approach involves the establishment of rules that constrain the action of each unit or position into paths consistent with those taken by others in the interdependent relationship. An important assumption in coordination by routine is that the set of rules need to be internally consistent, and this requires that the situations to which they apply must be relatively stable, repetitive, and few enough to permit matching of situations with the appropriate rules. The second approach, **mutual adjustment**, involves the transmission of new information during the process of action. March and Simon (1958) refer to this as “coordination by feedback”. The more variable and unpredictable the situation, the greater the reliance on coordination by mutual adjustment.

Gittell (2002) identified two other approaches, namely **team meetings** and **supervision**. **Team meetings** give participants the opportunity to coordinate tasks directly with one another. According to organization theory, they increase the performance of interdependent work processes by facilitating interaction among participants and are increasingly effective under conditions of high uncertainty. Supervisors, also known as boundary spanners, are individuals whose primary task is to integrate the work of other people.

Socio-technical coordination approaches include the team situation awareness model by Endsley and Jones (1997, 2001), and the joint activity model by Klein et al. (2005). The **team situation awareness** model conceptualizes how teams develop high levels of situation awareness (SA) across members and includes four crucial elements on which team SA is built. These include an understanding of what constitutes SA requirements in team settings, devices, and mechanisms that are important for achieving high levels of shared SA and the processes that effective teams use.

As is pictured in Fig. 2, the **joint activity** model (Klein et al., 2005) distinguishes three specific types of process phases that are required for effective coordination namely: 1) Criteria for joint activity, 2) Requirements for joint activity, and 3) Choreography of joint activity. The criteria for joint activity are that participants intend to work together (known as the basic compact) and their work has to be interdependent.

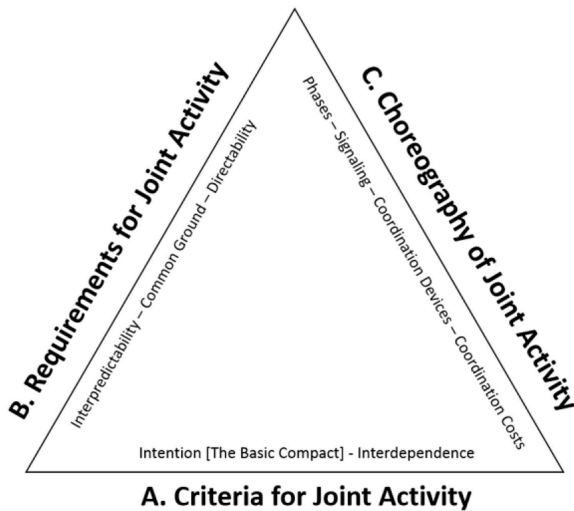


Fig. 2. Joint activity theory of Klein et al. (2005).

The basic compact constitutes a level of commitment for all parties to support the coordination process, e.g. the commitment to some degree of goal alignment, and commitment to try and detect and correct any loss of common ground that might disrupt the joint activity. If these criteria are satisfied, the parties have to fulfill certain requirements such as making their actions predictable, sustaining common ground, and being able to redirect each other. The form for achieving these requirements (the choreography) is a series of activities that are guided by various signals and coordination devices.

4. MAS supported AOC policy

Castro (2013) developed MASDIMA as a MAS based tool set in support of an AOC policy in which several human roles are automated. This tool set makes use of intelligent software agents that negotiate with each other to manage airline disruptions, but keeps the human in the loop as supervisor. This means that both software agents and humans have to coordinate their actions. Coordination approaches between software agents as well as between human agents are therefore relevant for a MAS supported AOC policy.

4.1. Architecture of MAS supported AOC policy

Fig. 3 compares current AOC with MASDIMA supported AOC. This figure shows how software agents replace the passenger team, aircraft team, and crew team. The roles for humans that remain in AOC are the supervisor, maintenance services, and flight dispatch.

Fig. 4 shows how software-agents form MASDIMA's architecture. Rectangles represent software agents and rounded rectangles represent human agents. Solid arrows show interactions between agents, whereas dashed arrows indicate querying of one of the many data sources. Data sources are shown as cylinders in the oval located on the lower side of the figure. The cloud around the supervisor agent and manager agents indicates that these agents negotiate with each other.

MASDIMA manages a disruption as follows. The first step in the disruption management process is to detect disruptive events. This is done by the monitor agent which reads the event database (see dashed arrow between the monitor agent and the data sources in Fig. 4). Flight dispatch is responsible for entering information about disruptions in the event database. From the event database, the monitor agent assesses which airlines resources are affected and requests a solution to the disruption problem from the supervisor agent. A problem has three dimensions: the aircraft, the crew, and the passenger dimension. To find a solution the supervisor agent negotiates with an aircraft, crew, and passenger manager. Each manager can solve one dimension of the

problem. To get a solution on all three dimensions, the manager agents negotiate with each other. I.e. the aircraft manager solves aircraft related problems, but for problems that also affect the crew it has to negotiate with the crew manager. The crew manager then provides a solution to the crew problem. This negotiation allows manager agents to present an integrated solution, that is, a solution to all dimensions of a disruption problem.

A manager agent has many options to solve a disruption. For instance, to solve an aircraft related problem, the aircraft manager may delay or cancel the flight, or exchange aircraft. To find which of these options is best, each manager defines its utility function. This allows the manager agents to find a solution that is optimal from their perspective. The aircraft manager aims to minimize aircraft delay and aircraft cost, the crew manager minimizes crew delay and crew cost, and the passenger manager minimizes passenger trip time and passenger costs. The utility function of each manager agent is defined in such a way that it aims to maximize its utility. E.g. the result of maximizing utility for the aircraft manager is that it minimizes aircraft delay and aircraft cost.

Equation (1) shows the utility function $U_{a/c}$ that is adopted for the aircraft manager in MASDIMA. This utility $U_{a/c}$ is a function of t_{ad} , $C_{a/c}$, ω_1 and ω_2 . t_{ad} is the total aircraft delay for all flights in the schedule, $\max(t_{ad})$ is the total aircraft delay of the solution that results in the maximum delay. Similarly, $C_{a/c}$ is the total aircraft operating cost and $\max(C_{a/c})$ is the aircraft cost for the solution with the maximum total aircraft operating cost. The weights ω_1 and ω_2 determine the relative importance of aircraft delay versus aircraft cost.

When the supervisor agent requests a solution from the manager agents, it may receive three different solutions. This is because the manager agents each have their own definition of what is an optimal solution. To determine what solution is best, the supervisor agent defines its utility function. This function covers all three dimensions of the problem. It takes into account aircraft cost and aircraft delay, crew cost and crew delay, passenger compensation cost and aircraft delay, crew cost and crew delay, passenger compensation cost and passenger travel time.

4.2. Negotiation protocol of MAS supported policy

MASDIMA uses the Generic Q-Negotiation (GQN) protocol that is used for multi-attribute negotiation with several rounds and qualitative feedback with interdependent dimensions. At the heart of this protocol are the notions of an Organizer Agent (OA) and Respondent Agents (RA) (see Fig. 5). The organizer agent is responsible for defining the problem, dividing it into sub problems, and initiating the negotiation as well as evaluating the proposals received from the respondent agents. The respondent agents have an expertise on their problem dimension and are entitled to negotiate with each other in order to present an integrated solution to the organizer agent. The organizer agent is the supervisor and the main respondent agents are the aircraft manager, crew manager, and passenger manager in Fig. 4. Negotiations are organized at two levels, i.e. the organizer/respondent agents level (Level 1), and the respondent agents level (Level 2). At level 1, the supervisor agent negotiates with the manager agents. At the second level, the manager agents negotiate with each other to find a solution to all three dimensions of the problem. These levels are described below.

4.2.1. Level 1: organizer/respondent agent negotiation

Upon identification of a problem at Level 1, the supervisor issues a call for proposals (CFP) to the three expert managers. Issuing a CFP triggers a negotiation process for which a state diagram is given in Fig. 6). The Level 1 negotiation states are:

- **State 1** – CPF has been sent by the supervisor to the three expert managers.
- **State 2** – **More than one proposals received:** The supervisor evaluates all proposals received from the managers and select the

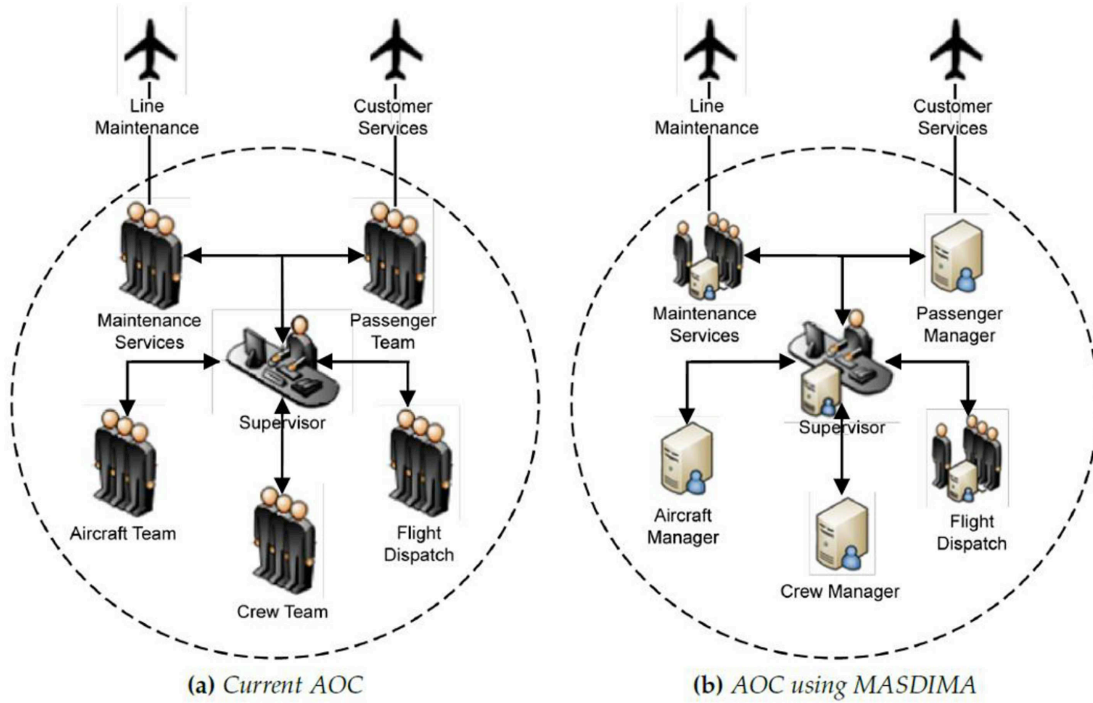


Fig. 3. Castro (2013) proposes to automate several of the roles in AOC.

winning solution. Hence the current negotiation round ends in state 5.

- **State 3 - Only one proposal received:** In state 3, the negotiation might end if the proposal is accepted by the supervisor. If the proposal is not accepted, a new CFP is sent, which brings the negotiation back to state 1.
- **State 4 - No proposals received:** In this case no solution has been identified for the problem. The supervisor can either change his preferences and issue a new CFP (state 1) or terminate the negotiation without a solution (Final).
- **State 5 – Negotiation round successfully finished:** This means that one of the proposals from state 2 remains, which is

subsequently moved to state 3.

It should be noted that the manager agents learn from the feedback provided by the supervisor. The state of an agent keeps a list of the feedback received from the supervisor in previous rounds. Based on its state, a manager may perform an action on an attribute. An action can prescribe to increase, decrease, or keep the value of an attribute. The reward affects the probability that an agent chooses a certain action in a next negotiation round.

4.2.2. Level 2: inter respondent agent negotiation

Each respondent agent needs to present a proposal with the solution

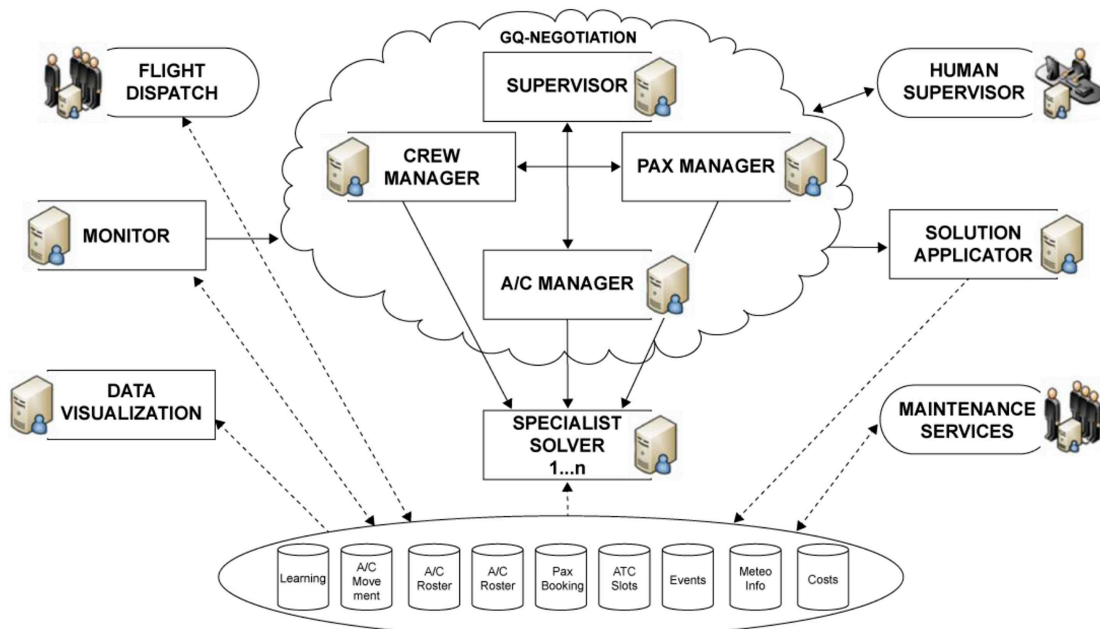


Fig. 4. MASDIMA architecture (Castro, 2013).

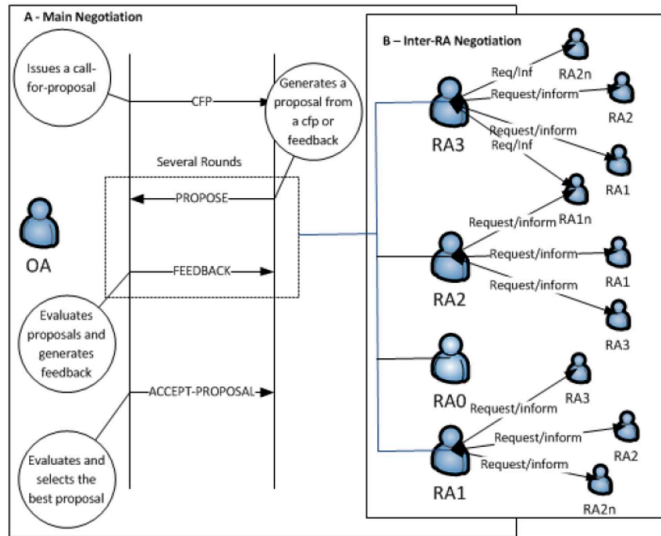


Fig. 5. GQN Agent Types, Roles, and Negotiations (Castro, 2013) - The figure shows one main (level 1) negotiation with several rounds and negotiations between the respondent agents in each round. The organizer agent assumes the role of the initiator of level 1 negotiations, whereas the respondent agents assume the role of participant. The respondent agents are initiators and participants in inter-RA (level 2) negotiation. If the respondent agent has the full expertise to present a proposal, it does not need to engage in a negotiation process with other respondent agents.

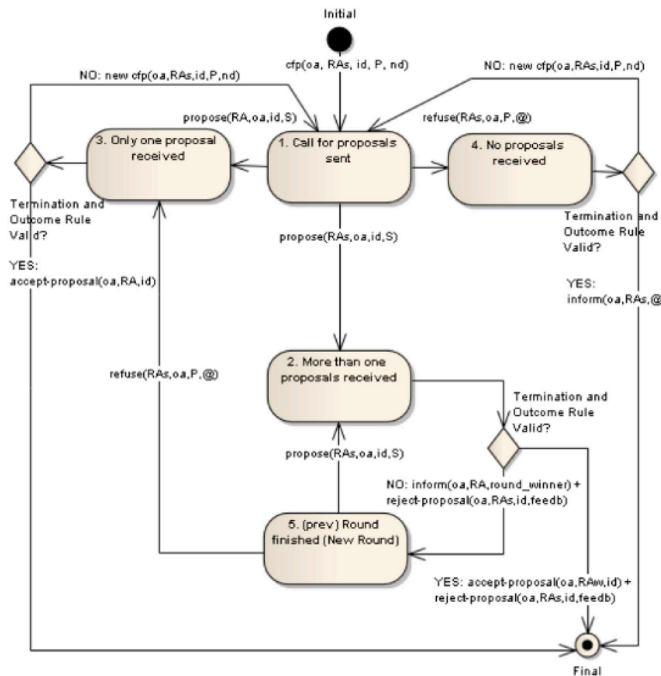


Fig. 6. Level 1 negotiation state diagram (Castro, 2013).

for the whole problem and not only for the dimension that corresponds to its competence. To achieve this, respondent agents negotiate with each other. Fig. 7 depicts the transition diagram for the negation process, including its states. The states are described below.

- **State 1 – Partial solution requests sent:** The respondent agent issues requests to all other respondent agents asking for partial solutions that can complement the candidate solution it found for its dimension. Such requests include restrictions that each respondent agent needs to consider before sending its partial solution. This

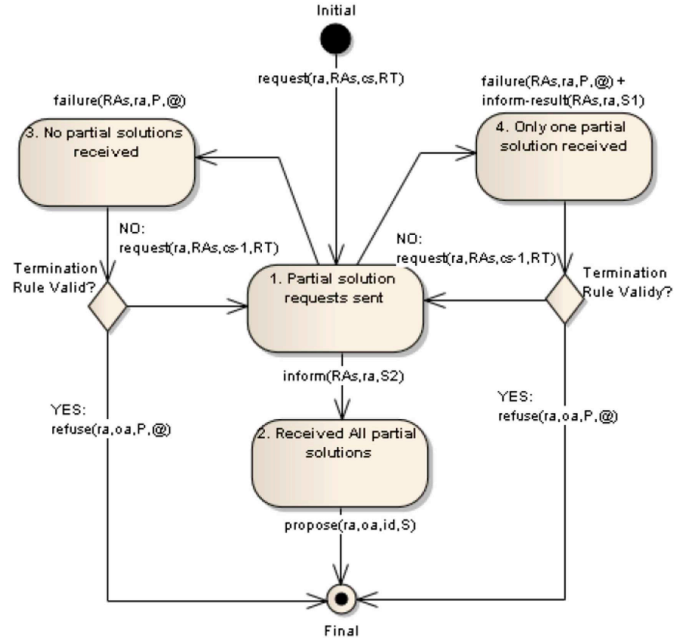


Fig. 7. Level 2 negotiation state diagram (Castro, 2013).

ensures that the combined candidate solution is valid. The issue of requests leads to state 1. From state 1, various states can be reached:

- **State 2 – Received all partial solutions.** In this case, all respondent agents present a partial solution to the respondent agent. The respondent agent ends the negotiation by issuing a proposal solution that covers all dimensions.
- **State 3 – No partial solutions received.** In this case, none of the respondent agents has a partial solution. The respondent agent can issue a new request, or terminate the negotiation. When the negotiation is terminated, the respondent agent informs the organizer agents that he is not able to make a proposal on that round of the main negotiation.
- **State 4 – Only one partial solution received.** In this case only one partial solution for a single dimension is received. The other respondent agents do not have a partial solution that complies with the request sent by the respondent agent. In this case, the respondent agent can issue a new request, or terminate the negotiation. When the negotiation is terminated, the respondent agent informs the organizer agents that he is not able to make a proposal on that round of the main negotiation.

5. Evaluating Aoc disruption management policies

This section compares the MAS supported AOC policy with the four human-team coordination policies considered by Bouarfa et al. (2016). First the AOC disruption management policies are outlined. Next the five policies are compared regarding their use of specific coordination techniques. Subsequently the benchmarking airline disruption scenario is described as well as the performance indicators used in the benchmarking. Next this benchmarking scenario is evaluated under the MAS supported policy. Finally, the MAS supported policy benchmark results are compared against those obtained for the four human coordination policies.

5.1. Five AOC disruption management policies

We number the five specific AOC disruption management policies to be considered as P1-P5. Policies P1-P3 are based on established AOC practices (Bruce, 2011a; 2011b). Policy P4 is based on the joint activity coordination theory of Klein et al. (2005). P5 is the MAS supported AOC

policy described in section 4. Policies P1–P4 are summarized below; for a complete description see (Bouarfa et al., 2016).

1) *AOC policy P1 – Elementary level of performance*: Airline controllers identify various basic level considerations such as aircraft patterns and availability, crew commitments and maintenance limitations. For example, when a maintenance problem is reported, controllers at this level appear to acknowledge the information provided and begin considering the basic consequences of the scenario. They also identify opportunities to replace the aircraft or rebook passengers on alternative flights.

2) *AOC policy P2 – Core level of performance*: Airline controllers have a greater comprehension of the problem. They take into account more complex consequences of the problem than those evident at the elementary level. Several constraints such as crew restrictions, slot times, and curfews are identified at this level. Controllers would for instance negotiate maintenance requirements and crew limitations in order to reduce the risk of breaching the curfew.

3) *AOC policy P3 – Advanced level of performance*: Airline controllers demonstrate thinking beyond the immediacy of the problem. They examine creative ways to manage the disruption. For instance, controllers at this level would consider more complex crewing alternatives such as positioning a crew from one airport to another airport where the flight crew is needed. Also, in case of a maintenance problem, controllers at this level would seek alternative information and recheck the reliability of information, e.g. through organizing a conference call with the maintenance watch people.

4) *AOC Policy P4 – Joint activity policy*: The fourth AOC policy P4 is based on the joint activity framework of Klein et al. (2005) that is depicted in Fig. 2. In order to apply the joint activity based approach to AOC disruption management, Bouarfa et al. (2016) have identified rules that AOC agents should adhere to in order to comply with Klein's joint activity theory (Klein et al., 2005). These rules are defined for each of the three types of joint activity process phases, and are described in more detail in Bouarfa et al. (2016). Subsequently, Bouarfa et al. (2016) have shown through ABMS the great potential this joint activity theory has for improving current AOC policy.

5.2. Coordination approaches of P1–P5

Table 1 shows which coordination approaches reviewed in Section 3 apply for each of the five policies P1–P5. This shows that almost all coordination approaches of Section 3 (except Voting methods) are used within one or more of the five AOC policies P1–P5.

The AOC policies P1–P4 have several coordination approaches from Section 3 in common, i.e. master/slave, contract net protocol, multi-agent planning, routines/protocols, mutual adjustment, supervision and criteria for joint activity. This commonality stems from the typical

Table 1
Approaches from the coordination literature used by AOC policies P1–P5.

Coordination Approach	Simulated Coordination Policies				
	P1	P2	P3	P4	P5
Master/Slave technique	+	+	+	+	+
Contract net protocol	+	+	+	+	+
Multi-agent planning	+	+	+	+	+
Negotiation protocol	–	+	–	–	+
Voting methods	–	–	–	–	–
Routines/protocols	+	+	+	+	–
Mutual adjustment	+	+	+	+	–
Supervision	+	+	+	+	–
Team meetings	–	–	+	+	–
Criteria for joint activity	+	+	+	+	–
Requirements for joint activity	–	–	–	+	–
Choreography of joint activity	–	–	–	+	–
Team Situation Awareness	–	–	–	+	–

airline manner of flight planning and their AOC organization. Policy P1 has only one coordination approach complementary to this common set, i.e. dedicated routines/protocols in resolving a disruption. Policy P2 also makes use of negotiation protocols between team members as a complementary approach. Policy P3 is similar to Policy P2, although it makes use of team meetings instead of negotiation protocols. Policy P4 is an extension of Policy P3 with Team Situation Awareness (Endsley and Jones, 1997, 2001) and a replacement of the dedicated routines/protocols of P3 by the rules of joint activity theory (Fig. 2). Policy P5 has less commonality with the P1–P4 because human-centred coordination policies play a lesser role.

5.3. Benchmark scenario

As has been explained in the introduction, the benchmark scenario used to evaluate the MAS supported AOC policy P5 originates from Bruce (2011b). Bouarfa et al. (2016) used the same scenario to evaluate human coordination policies P1–P4. The scenario considers an aircraft mechanical failure at an airline's outstation. As a consequence of this failure, the flight is delayed by 3 h. This may cause crew to exceed their legal duty time. AOC is to find and implement a solution.

Table 2 shows the flight schedule for the two long-haul aircraft that are in Europe around the time of the disruption. A_{dep} and A_{arr} refer to airport of departure and arrival, respectively. Reg. is the aircraft registration, and Day indicates a fictional day of departure. The scheduled time of departure and the scheduled time of arrival are represented by t_{sd} and t_{sa} . Furthermore, t_{flight} is the flight time in minutes (excluding taxiing), and d_{flight} the great circle distance in kilometers between departure and arrival airport. The airline's fictional hub Pacific (PCF) is assumed to be located 806 km and 90 flight-minutes south of Hong-Kong (Bouarfa et al., 2016).

The scenario chronology is as follows. The disruption is detected at 06:55 UTC. Flight 705 is unserviceable in Paris (CDG). An engineer observed a hydraulic pump leak during routine maintenance. The engineer contacted the ground-supervisor and informed him about the leak. Subsequently, the engineer found out that the hydraulic pump should be changed. Moreover, because of bad weather he believed it is best to do the pump change in a hangar. This would delay the flight by at least 3 h, such that the crew of flight 705 exceeds their legal duty time.

Flight 705 is not fully booked. It carries 420 passengers on an aircraft with a maximum seat capacity of 450 passengers. There is another long haul flight from Europe to Pacific around the same time. This is flight 701 from London (LHR) to PCF, carrying 445 out of 450 passengers. Because this flight is almost fully booked, it cannot carry any additional passengers from flight 705. There are no reserve aircraft available at either CDG or LHR.

An expert panel described the optimal solution to fly in reserve crew from PCF to BOM. Subsequently, this reserve crew can fly the aircraft from BOM to PCF without violating their legal duty time (Bruce, 2011b).

Table 2

Flight schedule for the two long-haul aircraft that are in Europe around the time of the disruption. Flight schedule from Bruce (2011b). Flight times and flight distances from Travelmath.com (2015).

Flight	A_{dep}	A_{arr}	Reg.	Day	t_{sd}	t_{sa}	t_{flight}	d_{flight}
700	PCF	LHR	LHA	1	09:10	22:05	771	10453
706	PCF	CDG	LHB	1	10:55	22:55	738	10413
701	LHR	PCF	LHA	2	08:35	21:30	771	10453
705	CDG	PCF	LHB	2	09:00	21:00	738	10413
761	PCF	YVR	LHA	3	05:00	13:45	816	11094
700	PCF	LHR	LHB	3	07:00	19:55	771	10453

5.4. AOC performance indicators

To assess the solution corresponding to the MAS supported policy, the same performance indicators used by Bouarfa et al. (2016) are considered:

- *Flight*: This performance indicator describes what happens to the flight. For instance, the flight can be cancelled, delayed, or diverted.
- *Integrated solution*: A solution to an airline disruption problem has three dimensions: the aircraft solution, the crew solution, and the passenger solution. For instance, the aircraft solution could be to fix the aircraft mechanical failure.
- *Disruption management time*: The time it takes to solve a disruption problem starts when the disruption is first detected, and ends when a solution is found and implemented by AOC. Implemented here means that the changes are applied to the operational schedules, and all involved parties are informed of the solution.
- *Operational Costs*: The cost model by Bouarfa et al. (2016) is used to compare the solution by policy P5 to the solutions corresponding to policies P1–P4. For the operational costs, Bouarfa et al. (2016) use recent cost data from Air France-KLM (Air France - KLM, 2014).
- *Passenger compensation*: European legislation prescribes that, under certain conditions, passengers can claim compensation from the airline if their flight is cancelled or delayed. When a flight is cancelled without prior notice, passengers flying longer than 3500 km can claim €600, or €300 if the airline provides an alternative itinerary (European Parliament, 2004).
- *Passenger delay*: The passenger delay is calculated as the difference in hours between the scheduled time of departure and the estimated time of departure.

5.5. Evaluation of policy P5 on the benchmark

The evaluation of the benchmark scenario under policy P5 has been established by following the description of the MAS support system by Castro (2013, sections 7.6–7.7). The details of this evaluation are described by Müller (2015, section 5.4). Here we provide an outline of what happens under P5 for the scenario considered.

First the ground engineer detects the leak and calls the ground-supervisor. The ground supervisor enters the disruption into the Aircraft Movement System, which can be read from the AOC center. Flight dispatch notices the disruption, and enters the information into MASDIMA's event database.

MASDIMA's monitor agent detects the disruptive event from the database and requests the supervisor agent to solve the problem. As the aircraft, crew, and passenger dimensions are all affected, all manager agents participate in the negotiation to propose a solution. For the aircraft manager agent, the optimal solution is to delay the flight until the aircraft is fixed. For the crew manager agent, the optimal solution is to exchange the crew from flight 706 with the crew from flight 705. The most optimal solution for the passenger problem is to keep the passengers on the same flight. The manager agents now start negotiating with each other according to the Generic Q-Negotiation protocol described in section 4.2. The outcome of this negotiation is an integrated solution covering all three dimensions and proposed to the supervisor.

This is also the solution proposed to the human supervisor. If the human supervisor accepts the proposal, the solution is implemented in the operational schedule.

Application of policy P5 to the benchmark scenario yields as solution to delay flight 705 in order to fix the aircraft mechanical problem, to replace the crew from flight 705 by the crew from inbound flight 706, and to keep passengers on the same flight. It is assumed that the human supervisor accepts this solution proposal.

The time to obtain the final solution was determined based on (Machado (2010), Castro and Oliveira (2011)) together with data from developer of policy P5. The timing of the different steps is presented in

Table 3

Timing of steps in disruption management policy P5.

Step	t_{solve} (min)
Ground engineer contacts station supervisor to inform him about the leak	3.5
Station supervisor enters information on the disruption in the AMS	4.5
Flight dispatch reads AMS and decides to put the information in the event database	2.5
Flight dispatch puts disruption in event database	4.5
MASDIMA solves the disruption	1
Human supervisor approves solution by MASDIMA	0.5
MASDIMA applies solution on the operational plan	0.5
Total time to resolve disruption	17

Table 3.

5.6. Results for P5 versus results for P1–P4

The P5 solution of the benchmark scenario is now compared to those obtained by Bouarfa et al. (2016) for P1–P4. First of all, the P5 solution appears to be equal to the solution of P4. However, this solution differs significantly from the solutions that have been obtained by P1–P3.

Table 4 shows the performance results obtained for the five AOC policies P1–P5 on the challenging benchmark scenario. The results for P1–P4 are from Bouarfa et al. (2016).

The results in Table 4 show that for P5, the airline operating cost was found to be €326,000 and zero cost for legal pax compensation. These are equal to the costs that were identified for P4, but they are significantly better than those for the conventional AOC policies P1–P3.

Table 4 also shows a small complementary advantage of policy P5 over P4. P5 needs about 20% less time than policy P4 to find and implement the same resolution for the challenging airline disruption scenario. However, in comparison with the conventional policies P1–P3, the MAS supported AOC policy P5 significantly reduces the time needed to find and implement its solution.

6. Expert-based evaluation of P5

Introducing MAS supported policy P5 to AOC might have an impact on the current airline disruption management process. To investigate this, an AOC expert from a major European airline was interviewed. This has led to the identification of several key AOC tasks that could be affected. These tasks have been evaluated from a current AOC perspective.

6.1. AOC tasks considered

The AOC tasks have been identified during an interview with an AOC expert. This interview had the form of an open discussion on the operational changes implied by MAS supported policy P5. The list of tasks identified was not exhaustive, but it identified a rough cross-section of AOC tasks affected by policy P5 (See Table 5).

The introduction of the MAS supported policy P5 may change the workload for people that remain in AOC. Fig. 3 compared AOC under policies P1–P4 (Fig. 3a) to AOC under the MAS supported policy P5 (Fig. 3b). This shows that the passenger team, the aircraft team, and the crew team are replaced by the automated passenger manager, the automated aircraft manager, and the automated crew manager respectively. In order to gain insights into potential effects of these replacements, each task will be evaluated in relation to workload.

6.2. Results of expert-based task evaluation

The expert-based evaluation identified potential problems for each

Table 4

Results of the scenario-based analysis. The novel policy P5 finds the same solution as policy P4, which is better than the solutions found by policies P1–P3. Results for policies P1 to P4 originate from Bouarfa et al. (2016).

AOC policy	Flight	Aircraft mechanical problem	Crew problem	Passengers problem	Minimum disruption mgmt time	Costs for the airline [Euros]		Passenger travel delay (hours)
						Operating costs	Legal pax. compensation	
P1	Cancelled	Fixed	Not resolved	Pax. accommodated in hotel (i.e. distressed)	26 min	326 kEUR	168kEUR	24
P2	Cancelled	Fixed	Not resolved	Pax. accommodated in hotel (i.e. distressed)	30 min	326 kEUR	168 kEUR	24
P3	Diverted	Fixed	Resolved	Pax. significantly delayed due to fixing aircraft and diverting	33 min	360 kEUR	126 kEUR	8
P4	Delayed	Fixed	Resolved	Pax. delayed until aircraft is fixed	20 min	326 kEUR	0 kEUR	3
P5	Delayed	Fixed	Resolved	Pax delayed until aircraft is fixed	17 min	326 kEUR	0 kEUR	3

Table 5

AOC tasks affected by MAS supported policy P5.

AOC Task #	Task Description
1	Monitoring flight safety
2	Monitoring flight slots
3	Judging the number of pilots required to operate the flight
4	Checking whether crewmembers reported for duty
5	Finding a reserve crewmember
6	Finding a new itinerary for disrupted passengers
7	Asking for a crewmember's permission

of the tasks in Table 5. These problems are described below for each of the seven tasks.

6.2.1. Monitoring flight safety

Monitoring flight safety is currently the task of flight dispatch. Castro (2013) proposes that flight dispatch's responsibilities should also include uploading information on disruptions to the MAS based tool (e.g. weather updates). This task might lead to an increased workload and hence could interfere with monitoring flight safety.

6.2.2. Monitoring flight slots

Flight slots are used to manage capacity and are provided to operators up to 2 h before departure. These slots specify a time for the aircraft to enter a specific airspace or depart from an airport. Currently, flight dispatch follow flights² and monitor the assigned slots. The MAS based tool does not automatically monitor slots, and therefore lacks an important functionality.

6.2.3. Judging the required number of pilots

Most flights require two pilots for operation: a captain and a first officer. In some cases however, a relief pilot is required. For instance, in case of a long haul flight exceeding duty time, or adverse weather at arrival airport causing delay. Currently flight dispatchers judge if a third pilot is required based on last minute information. If this task is automated, it could lead to not having enough pilots to operate the flight.

6.2.4. Checking if crewmembers reported for duty

Crewmembers generally have to report for duty by checking-in. The Crew Tracking System (CTS) keeps track of all crew check-ins. Currently, flight dispatchers monitor the CTS to check if all crewmembers reported for duty and take necessary action in case of missing crew. There is no feature in the MAS supported tool that automatically extracts information from CTS and notifies the supervisor in cases of missing crew.

6.2.5. Finding a reserve crewmember

Finding a replacement crew is necessary in cases where

crewmembers cannot report for duty. E.g. due to illness or other circumstances. Currently, it is the task of the AOC crew team to find an alternative crew. This task usually involves checking the crew reserve list, identifying the most optimal crew (e.g. in terms of cost, distance, and experience), and contacting the crew either by phone or text message. Upon introducing the MAS supported policy P5, this task would shift to the AOC supervisor and would significantly increase his workload.

6.2.6. New itineraries for disrupted passengers

Airlines are usually required to provide alternative itineraries to disrupted passengers whose flight was cancelled. In current AOC organizations, this role is usually performed by the passenger team who use various decision support tools to find the best alternatives for the passengers. If this task is automated using policy P5, the proposed solutions still need to be reviewed and approved by the AOC supervisor. This will lead to a significant increase of the supervisor's workload.

6.2.7. Asking for a crewmember's permission

During disruptions, the AOC crew team might need additional crewmembers than the ones available in the reserve list. For such a situation, crew controllers need to request permission from crewmembers, and in some cases negotiate with them to operate specific flights. Since the MAS supported policy P5 eliminates the crew team, this role is shifted towards the AOC supervisor which would lead to an increase in his workload.

6.3. Challenging scenarios

In addition to the expert-based evaluation of specific AOC tasks, the expert also identified the following four challenging scenarios.

6.3.1. Finding solutions to exceptional situations

There are many examples in which airline employees bypass standard procedures to solve a situation that would otherwise have resulted in a disruption. If an airline were to use MAS supported policy P5, the experience and ability to judge such situations has to be centered in the supervisor. Introducing policy P5 therefore increases the task of the supervisor in handling exceptional situations by for example circumventing standard procedures.

6.3.2. Getting passengers nearer to their destination

In some cases it is not possible to get passengers to their final destination. As an alternative, an airline may choose to transport passengers as close as possible to their destination. For instance when passengers cannot directly fly to New York (JFK) from Amsterdam (AMS), the airline may still be able to fly passengers to Philadelphia (PHL) and transport passengers by another transport mode to New York. Currently the passenger team is responsible for finding these types of solutions. There is no feature in policy P5 proposing such solutions.

6.3.3. Protecting the airline's strategic position

Some routes are important to an airline for strategic reasons, whilst flights on these routes are relatively easy or cheap to cancel. From literature it appears that no one in AOC is currently explicitly responsible for protecting the airline's strategic position, but it is assumed that this is inherent to AOC tasks. MAS supported policy P5 does not include a feature that protects the airline's strategic position because it replaces several human AOC roles. Such task will shift to the remaining people and increases their workload.

6.3.4. Taking into account political situations

Political sensitivities between countries may require attention from an airline, especially when the airline flies to disputed areas. Currently no one in AOC is explicitly responsible for this task, but it is assumed to be inherent to AOC roles. MAS supported policy P5 does not include a feature to check for politically sensitive situations. Ensuring the political neutrality of an airline therefore shifts towards the people that remain in AOC and increases their workload.

6.4. Main findings

The expert-based task analysis of the MAS supported AOC policy identified three categories of causes for potential problems.

First, for several tasks the workload increases for the supervisor, mainly because he/she is the human that remains in AOC after the introduction of the MAS supported AOC policy.

Second, eliminating the aircraft team, the crew team, and passenger team asks for centring their experience with the AOC supervisor. The analysis showed that the MAS decision-support tool provides a range of solutions to standard disruption problems, such as finding a reserve crewmember or an alternative itinerary. Because the MAS support eliminates part of human AOC tasks by software agents, AOC loses on human flexibility. This may have a negative effect for cases where pre-programmed solutions are not the best options.

Third, collaboration is key to finding solutions to disruption problems. Automating several human AOC roles also means their contacts and their ability to collaborate are lost. This may reduce AOC's resilience in recovering from disruptions and may increase the workload for the people that remain in AOC.

7. Conclusion

This paper evaluated an AOC disruption management policy that makes explicit use of the MAS-based decision support system MASDIMA. Part of this evaluation was to make a comparison with the four human-based AOC disruption management policies that have been evaluated by Bouarfa et al. (2016). Three complementary types of evaluations have been conducted.

Firstly, a comparison of the five policies on the coordination techniques used, show that a MAS supported AOC policy makes use of fewer coordination techniques than human-based AOC policies do. The four core coordination techniques that remain are Master/slave, Contract net protocol, Multi-agent planning and Negotiation protocol (see Section 5.2).

Secondly, the performance benchmarking by Bouarfa et al. (2016) of four human-based AOC policies has been extended with a performance benchmarking of the MAS supported AOC policy. This benchmarking compares performance of each of the five policies on a challenging airline disruption example from Bruce (2011b). The benchmarking results show that the MAS supported AOC policy performs as well as an advanced human-based AOC policy, but does significantly better than current AOC policies. A small complementary advantage of the MAS supported AOC policy is that it needs less time than human-based AOC policies.

Based on the benchmarking results obtained for the challenging airline disruption scenario alone, it can be concluded that MAS-based

decision-support has great potential in improving established AOC disruption management policies. Complementary expert-based task analysis showed that there are some remaining issues for which further improvements of MAS supported AOC policy might be beneficial.

There are valuable directions for follow-up research. One direction for follow-up research is to extend the benchmarking of five airline disruption management policies with one or more other AOC policies. Another direction for follow-up research is to develop another good benchmark scenario, followed by an evaluation of the application of the five AOC policies to this airline disruption scenario. A possible source for the development of another benchmark scenario is to incorporate potential criticalities that have been identified in Section 6 for the remaining human tasks in the MAS supported policy. A third direction for follow-up research is to extend the expert-based task analysis work by interviewing other AOC experts from various airlines.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jairtraman.2018.05.009>.

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